

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 50%

QUESTIONS: 30

Time Duration: 30 Minutes
Total Marks:30

Annexure A

MCQ

Code of Conduct:

I A G Jayapradhan (192471016) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: A G Jayapradhan

Annexure A

Circle the most appropriate option for each question.

Q1. Which of the following best describes a Logical Agent in AI?

- | | |
|--|---|
| a) An agent that reacts randomly to the environment | c) An agent that only follows pre-defined scripts |
| b) An agent that uses a knowledge base and logical inference to decide actions | d) An agent that learns from rewards |

Q2. In Propositional Logic, ' $P \rightarrow Q$ ' is logically equivalent to:

- | | |
|----------------------|-------------------------------------|
| a) $P \wedge \neg Q$ | c) $\neg Q \rightarrow \neg P$ only |
| b) $\neg P \vee Q$ | d) Both b and c |

Q3. Conjunctive Normal Form (CNF) requires the formula to be expressed as:

- | | |
|--|-----------------------------------|
| a) A disjunction of conjunctions | c) A conjunction of literals only |
| b) A conjunction of disjunctions (clauses) | d) A sequence of implications |

Q4. In FOL, the sentence $\forall x: \text{Student}(x) \rightarrow \text{Passes}(x)$ means:

- | | |
|--|---|
| a) Some students pass | c) If something passes, it is a student |
| b) Every entity that is a student passes | d) There exists at least one student who passes |

Q5. Unification in FOL is the process of finding substitutions that:

- | | |
|---|---|
| a) Convert clauses to CNF | c) Prove a goal using backward chaining |
| b) Make two logical expressions syntactically identical | d) Derive facts using Modus Ponens |

Q6. In the Resolution algorithm, deriving the empty clause $\square$ indicates:

- | | |
|-------------------------------------|--|
| a) The knowledge base is consistent | c) A contradiction — the original goal is proven |
| b) The negated goal is satisfiable | d) More clauses need to be added |

Q7. Forward Chaining is best described as a _____ strategy.

- | | |
|----------------------------|--------------------------------|
| a) Goal-driven / top-down | c) Heuristic-based depth-first |
| b) Data-driven / bottom-up | d) Probabilistic inference |

Q8. Backward Chaining begins reasoning from:

- | | |
|---|--|
| a) All known facts toward a conclusion | c) A random clause in the knowledge base |
| b) The goal and works backward to verify supporting facts | d) The resolution refutation tree |

Q9. Which step is NOT part of converting a sentence to CNF?

- | | |
|--|--|
| a) <i>Eliminate biconditionals</i> ($\leftrightarrow$) | c) Apply Skolemization to remove existential quantifiers |
| b) Move negations inward using De Morgan's laws | d) Distribute AND over OR |

Q10. Knowledge Engineering in FOL involves:

- | | |
|--|---|
| a) Training a neural network on domain-specific data | c) Writing heuristic search strategies |
| b) Defining domain concepts, relations, and axioms formally in FOL | d) Encoding reward functions for reinforcement learning |

Q11. *Modus Ponens states: If $P \rightarrow Q$ is known and P is true, then:*

- | | |
|---------------------|---------------------------|
| a) $\neg Q$ is true | c) P is uncertain |
| b) Q must be true | d) The KB is inconsistent |

Q12. The Most General Unifier (MGU) is the substitution that:

- | | |
|--|---|
| a) Is the most specific possible match | c) Eliminates all variables from a clause |
| b) Makes two terms identical with minimal variable binding | d) Applies resolution to produce CNF |

Q13. Which inference method is more appropriate when a specific goal is known?

- | | |
|-----------------------|----------------------|
| a) Forward Chaining | c) Backward Chaining |
| b) Truth Table method | d) A* search |

Q14. Propositional Logic differs from First Order Logic because it:

- | | |
|---|-------------------------------|
| a) Supports probabilistic reasoning | c) Is always more expressive |
| b) Cannot represent objects, relations, or quantifiers — only truth values of atomic propositions | d) Handles uncertainty better |

Q15. Resolution Refutation proves a goal G by:

- | | |
|---|--------------------------------------|
| a) Directly deriving G from the KB using forward chaining | c) Checking G in the truth table |
| b) Adding $\neg G$ to the KB and deriving a contradiction (empty clause $\square$) | d) Applying Modus Tollens repeatedly |

Q16. Learning from observation in AI refers to:

- | | |
|--|---|
| a) An agent receiving numerical rewards from the environment | c) Acquiring knowledge by following explicit symbolic rules |
|--|---|

b) Improving performance by examining examples or experience from the environment

d) Memorizing a fixed dataset without generalization

Q17. Inductive learning is best described as:

a) Deriving specific conclusions from general rules

b) Generalizing a hypothesis from specific training examples

c) Memorizing training data exactly

d) Applying search algorithms to find optimal policies

Q18. In a Decision Tree, what does each internal node represent?

a) A class label / output decision

b) A probability distribution over outcomes

c) A test on an attribute / feature

d) A training data sample

Q19. Explanation-Based Learning (EBL) differs from inductive learning because it:

a) Requires large amounts of training data to generalize

b) Learns only from numeric data using regression

c) Uses prior domain knowledge to explain and generalize a single example

d) Applies unsupervised clustering to find hidden patterns

Q20. In statistical learning, a Naive Bayes classifier assumes:

a) All features are conditionally dependent on each other given the class

b) The data must follow a uniform distribution

c) All features are conditionally independent given the class label

d) Labels are determined by a reward function

Q21. Learning with complete data (no hidden variables) in a Bayesian network involves:

a) Using the EM algorithm to estimate missing values

b) Training a deep neural network with dropout regularization

c) Directly computing maximum likelihood estimates from observed data

d) Applying reinforcement signals to update weights

Q22. The Expectation-Maximization (EM) algorithm is used when:

a) Training data is fully labeled and no variables are hidden

b) An agent explores an unknown environment using trial and error

c) The dataset contains hidden (latent) variables that cannot be directly observed

d) Backpropagation fails to converge in a neural network

Q23. In a Neural Network, the activation function serves to:

a) Initialize the weights of each neuron to zero

b) Normalize the input dataset before training

c) Introduce non-linearity to enable learning of complex patterns

d) Store previously learned examples in memory

Q24. Which of the following correctly describes backpropagation in a neural network?

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|--|---|
| a) Weights are updated in the forward pass using input gradients | c) The error is propagated backward from output to input layers to adjust weights |
| b) Outputs of each layer are stored and reused in later epochs | d) A separate model is trained for each hidden layer independently |

Q25. A Support Vector Machine (SVM) — a type of kernel machine — classifies data by:

- | | |
|---|---|
| a) Minimizing the number of training examples used | c) Finding the maximum-margin hyperplane that separates classes |
| b) Grouping similar data points into clusters using centroids | d) Updating weights using Q-values from an environment |

Q26. The kernel trick in kernel machines (SVMs) allows:

- | | |
|--|---|
| a) Reducing the number of support vectors needed | c) Operating in a high-dimensional feature space without explicitly computing coordinates |
| b) Accelerating gradient descent convergence | d) Replacing hidden layers in a deep neural network |

Q27. In Reinforcement Learning, the agent aims to maximize:

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|---|--|
| a) Classification accuracy on a fixed test set | c) Cumulative reward received over time through interaction with the environment |
| b) The number of features selected from the training data | d) The size of the neural network hidden layer |

Q28. The Q-Learning algorithm in Reinforcement Learning is classified as:

- | | |
|---|---|
| a) A supervised learning algorithm requiring labeled datasets | c) A model-free, off-policy temporal difference learning method |
| b) An unsupervised clustering technique using centroids | d) A decision-tree-based ensemble classifier |

Q29. Overfitting in machine learning occurs when:

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|---|--|
| a) The model performs poorly on both training and test data | c) The model fits training data too closely and generalizes poorly to new data |
| b) The learning rate is set too low during gradient descent | d) The training set has more features than training examples |

Q30. Which of the following best distinguishes Reinforcement Learning from Supervised Learning?

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|---|--|
| a) RL uses labeled input-output pairs; SL uses reward signals | c) RL learns via trial-and-error with reward feedback; SL learns from labeled examples |
|---|--|

b) Both RL and SL require explicit reward functions

d) RL is always faster to converge than SL on any dataset

ANSWER

Q.No Answer

- Q1 b) An agent that uses a knowledge base and logical inference to decide actions
- Q2 d) Both b and c
- Q3 b) A conjunction of disjunctions (clauses)
- Q4 b) Every entity that is a student passes
- Q5 b) Make two logical expressions syntactically identical
- Q6 c) A contradiction — the original goal is proven
- Q7 b) Data-driven / bottom-up
- Q8 b) The goal and works backward to verify supporting facts
- Q9 d) Distribute AND over OR (*The standard CNF conversion distributes **OR over AND**, not AND over OR.*)
- Q10 b) Defining domain concepts, relations, and axioms formally in FOL
- Q11 b) Q must be true
- Q12 b) Makes two terms identical with minimal variable binding
- Q13 c) Backward Chaining
- Q14 b) Cannot represent objects, relations, or quantifiers—only truth values of atomic propositions
- Q15 b) Adding $\neg G$ to the KB and deriving a contradiction ($\square$)
- Q16 b) Improving performance by examining examples or experience from the environment
- Q17 b) Generalizing a hypothesis from specific training examples
- Q18 c) A test on an attribute/feature
- Q19 c) Uses prior domain knowledge to explain and generalize a single example
- Q20 c) All features are conditionally independent given the class label
- Q21 c) Directly computing maximum likelihood estimates from observed data
- Q22 c) The dataset contains hidden (latent) variables that cannot be directly observed
- Q23 c) Introduce non-linearity to enable learning of complex patterns
- Q24 c) The error is propagated backward from output to input layers to adjust weights
- Q25 c) Finding the maximum-margin hyperplane that separates classes
- Q26 c) Operating in a high-dimensional feature space without explicitly computing coordinates
- Q27 c) Cumulative reward received over time through interaction with the environment
- Q28 c) A model-free, off-policy temporal difference learning method
- Q29 c) The model fits training data too closely and generalizes poorly to new data
- Q30 c) RL learns via trial-and-error with reward feedback; SL learns from labeled examples

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions